

ATPL EASA STUDY GUIDE | 010 - AIR LAW

Chapter 21: Obstacles & Rescue Services (RFFS)

Syllabus Source: CAE Oxford (pp. 447-460) | Pass Target: >= 90% AviationExam

1. RESCUE AND FIRE FIGHTING SERVICES (RFFS CATEGORIES 1 TO 10)

RFFS category is determined by the overall length of the longest aeroplane using the aerodrome and its maximum fuselage width:

RFFS Category	Aircraft Overall Length	Max Fuselage Width	Representative Types
Category 1	0 m up to 9 m	2 m	Light singles (C152, PA-28)
Category 5	24 m up to 28 m	4 m	Regional turboprops (ATR-42)
Category 6	28 m up to 39 m	5 m	Regional jets (CRJ-900, E190)
Category 7	39 m up to 49 m	5 m	Single-aisle airliners (A320, B737-800)
Category 8	49 m up to 61 m	7 m	Wide-body twin-aisle (B767, B787, A330)
Category 9	61 m up to 76 m	8 m	Heavy airliners (B777, B747, A350)
Category 10	76 m up to 90 m	8 m	Very heavy airliners (Airbus A380-800)

[!] EXAM TRAP (AVIATIONEXAM) - RFFS Operational Response Time (AviationExam Core)

- Operational Objective: The operational objective of the RFFS is to achieve a response time of TWO (2) MINUTES to any point of each operational runway, and not exceeding THREE (3) MINUTES to any other part of the movement area. - Response Time Definition: The time between the initial call to the RFFS and the first responding vehicle(s) applying foam at the required rate.

2. EXTINGUISHING AGENT QUANTITIES & DISCHARGE RATES

Annex 14 specifies minimum amounts of usable water for foam production and discharge rates: - Category 6: 7,900 liters of water (Level B foam), discharge rate 4,000 L/min, complementary agent 225 kg. - Category 7: 12,100 liters of water (Level B foam), discharge rate 5,300 L/min, complementary agent 225 kg. - Category 9: 24,300 liters of water (Level B foam), discharge rate 9,000 L/min, complementary agent 450 kg.

3. OBSTACLE MARKING & LIGHTING STANDARDS

- **Marking Colors:** Fixed obstacles must be colored in a chequered pattern (orange/white or red/white) or alternating contrasting horizontal bands (at least 7 bands of orange/white or red/white).
- **Low-Intensity Lights:** Fixed red lights used for obstacles with height less than 45 m.
- **Medium-Intensity Lights:** Flashing red or flashing white lights used for obstacles between 45 m and 150 m.
- **High-Intensity Lights:** Flashing white lights (up to 200,000 cd day, 2,000 cd night) used for obstacles exceeding 150 m height (e.g. broadcast masts, tall cooling towers).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the maximum acceptable RFFS response time to any point of an operational runway in optimum conditions?

- [A]: 1 minute
- [B]: 2 minutes (not exceeding 3 minutes to other movement areas)
- [C]: 4 minutes
- [D]: 5 minutes

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The operational objective is 2 minutes to runway points, and max 3 minutes to any movement area.

Q2: Which RFFS category applies to an Airbus A320 with an overall length of 37.6 m and fuselage width of 3.95 m?

- [A]: Category 5
- [B]: Category 6
- [C]: Category 7
- [D]: Category 8

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Length 28 to 39 m with width up to 5 m corresponds to RFFS Category 6 (B737-800 is Cat 7 if length exceeds 39 m).

Q3: What type of lighting is required on obstacles exceeding 150 m in height?

- [A]: Low-intensity fixed red lights
- [B]: Medium-intensity flashing red lights
- [C]: High-intensity flashing white lights
- [D]: Green strobe lights

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Annex 14 mandates high-intensity flashing white lights for obstacles taller than 150 m.

Q4: What is the principal extinguishing agent required for aerodrome rescue and fire fighting?

- [A]: CO2 gas
- [B]: Dry chemical powder
- [C]: Water with foam-producing chemical (AFFF)
- [D]: Halon 1301

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Foam produced from water and film-forming fluoroprotein or aqueous film-forming foam (AFFF) is the primary agent.

Q5: What minimum amount of complementary dry chemical powder is required for RFFS Category 7?

- [A]: 100 kg
- [B]: 225 kg
- [C]: 450 kg

- [D]: 900 kg

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under Annex 14, Category 6 and 7 aerodromes require at least 225 kg of complementary dry chemical powder or gaseous agent.

Q6: What visual markings are required on service vehicles authorized to operate on aerodrome movement area

- [A]: Conspicuous single yellow or orange color, or display flashing yellow beacons
- [B]: Plain white with red stripes
- [C]: Military camouflage
- [D]: Blue flashing lights only

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. Movement area service vehicles must be colored conspicuously (yellow/orange) and display flashing yellow obstacle beacons.